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PLANETVORTEX — A Research Monograph

The Solar System as a $PSL(2,7)$ Figure: Exact Heptagonal Geometry, the Vortex Lattice and the Newtonian Planetary Registers

Mini-repository planetvortex/ of the TRIVORTEX framework — version 1.0.0. Every number quoted below is bound to a committed run protocol in results/protocols/; the tables quote recorded values, the proofs reference registered tolerances, and the figures are the output of the protocol-bound factory.

Abstract

This monograph develops the planetary gravity–geometry bench: the construction of a geometric figure — a regular heptagon with the Sun at the centre and the seven wandering planets at its vertices, with the seven Fano lines $\{i, i+1, i+3\} \pmod{7}$ drawn on it — whose symmetry group is $PSL(2,7)$, the smallest non-abelian simple group, and whose exact dimensions are computed in closed form and bound to the gravitational registers of the solar system. Three theorems and two lemmas organize the payload. Theorem A computes the exact dimensions of the figure: every Fano line is the congruent triangle with angles $(\pi/7, 2\pi/7, 4\pi/7)$ and area $\sqrt{7}/4 R^2$, by the classical product identity $\sin(\pi/7)\sin(2\pi/7)\sin(3\pi/7) = \sqrt{7}/8$. Theorem B shows that the seven three-vortex Trivortex cells hosted by the lines carry exactly equal Hamiltonians $-(\Gamma^2/2\pi)\ln(\sqrt{7}R^3)$ and congruent shape cycles — the dynamical shadow of $PSL(2,7)$, measured with spread 0.0. Theorem C certifies the heptagon lattice as a Havelock-stable relative equilibrium at $\omega_7 = 3\Gamma/2\pi R^2$ — the $N=7$ row of the sibling bench’s scan, cross-pinned bit-for-bit. Lemma D certifies Kepler’s third law dynamically across the eight planets — the mass-corrected periods match to 6.2×10^{-13} while the uncorrected formula misses by the physical signature $m_i/2M_\odot$ up to 4.77×10^{-4} (Jupiter). Lemma E measures the honest distance between the figure and the sky: the gravity ladder $\log_{10}(GM_i/GM_\oplus)$ summed over the seven lines spans 6.71 dex, and a mild gravity-weighted circulation ladder deforms the vortex figure by 0.30 within one rotation while the equal-circulation ring stays rigid to 9.8×10^{-15} . The bench runs the full Newtonian Sun + 8-planets simulation (energy

drift 2.1×10^{-13} over 12 years) and registers everything in a seven-stage P-ladder of committed JSON protocols.

1. Scope and provenance

The parent framework anchors on Theorem 3.1 — the closed-form choreography of three point vortices with the Chaplygin topological integral — and verifies the classical layer of its model with the V1–V4 ladder. The sibling bench polyvortex lifted the picture to N equal vortices and located the Havelock stability boundary exactly between $N=7$ and $N=8$. This mini-repository takes the $N=7$ object — the heptagon, the last stable ring — and asks what it becomes when its vertices are labeled by the seven wandering planets of the solar system and its centre by the Sun.

The provenance discipline is the framework’s own: a committed data table (the NASA register of Section 4), a committed structure (the Fano plane of Section 3), tolerances registered before the recorded runs, deterministic integration, and one JSON protocol per stage — seven stages, P1 through P7. Nothing in this volume quotes a number that no protocol carries. The mini-repository never imports the parent framework or the sibling bench at runtime; the sibling ladder appears only in the test suite as the reference oracle.

The units are astronomical: length in AU (exactly 149 597 870 700 m), time in Julian years, mass in solar masses; $G M_{\odot} = 4 \pi^2$ in these units by construction, and the planetary masses come from the committed GM register as $m_i = G M_i / G M_{\odot}$.

2. The two layers

The bench keeps two layers of the model honestly separated, exactly as the parent and the sibling do.

Layer N + P — the Newtonian planetary dynamics (numeric). The planets are point masses in the planar heliocentric idealization; the equations are the full pairwise Newtonian system

$$\ddot{r}_i = \sum_{j \neq i} 4 \pi^2 m_j \frac{r_j - r_i}{|r_j - r_i|^3}, i=0, \dots, 8,$$

integrated by classical RK4. The registers are the conservation of energy and angular momentum, the osculating elements (a, e) against the committed table, Kepler’s third law measured from the integrated motion, and the classical perturbation hierarchy.

Layer F + V — the exact figure and its vortices (analytic + numeric). The figure is the regular heptagon of circumradius R with the Fano plane on it; its dimensions are closed forms (Theorem A). The vortices are the Kirchhoff point-vortex flow

$$\dot{x}_i = -\frac{1}{2\pi} \sum_{j \neq i} \Gamma_j \frac{y_i - y_j}{r_{ij}^2}, \dot{y}_i = \frac{+1}{2\pi} \sum_{j \neq i} \Gamma_j \frac{x_i - x_j}{r_{ij}^2},$$

with the invariants $H = -\sum_{i < j} \Gamma_i \Gamma_j \ln r_{ij} / 2\pi$, $P = \sum \Gamma_i x_i$, $Q = \sum \Gamma_i y_i$, $I = \sum \Gamma_i |r_i|^2$. The connection between the layers is the gravity ladder $\delta_i = \log_{10}(G M_i / G M_{\oplus})$ carried by the vertices — and, honestly, the deficit of that connection (Lemma E).

3. The structure and the exact figure (Theorem A)

3.1 The Fano plane in two models

The Fano plane $PG(2, 2)$ is the unique projective plane on seven points: seven lines, three points per line, every pair of points on exactly one line. The bench builds it twice.

Model Z (cyclic). The points are the residues $Z_7 = \{0, \dots, 6\}$ — the heptagon vertices in orbital order. The lines are the seven triples

$$L_i = \{i, i+1, i+3\} \pmod{7}, i=0, \dots, 6.$$

Model B (binary). The points are the seven nonzero vectors of F_2^3 ; the lines are the triples $\{u, v, u+v\}$; the automorphism group is $GL(3, 2)$.

The automorphism group of either model has order 168 and is isomorphic to $PSL(2, 7)$ — the smallest non-abelian simple group, the group of orientation-preserving automorphisms of the Klein quartic. Stage P4 certifies, exhaustively: 168 line-preserving permutations of Z_7 (closed, all inverses present, transitive on points, stabilizers of orders $168/7=24$ both on points and on lines, the pair axiom on all 21 pairs); 168 invertible 3×3 matrices over F_2 ; and an explicit line-preserving bijection $\varphi: Z_7 \rightarrow F_2^3$ whose conjugation $\varphi \sigma \varphi^{-1}$ maps the cyclic group copy onto the $GL(3, 2)$ copy inside S_7 — the two models are one structure, and the certificate is constructive, not existential.

3.2 The exact dimensions

Theorem A. Draw the Fano plane on a regular heptagon of circumradius R by the triples $\{i, i+1, i+3\}$. Then every line is a scalene triangle with interior angles $(\pi/7, 2\pi/7, 4\pi/7)$, with sides $s_1 = 2R \sin(\pi/7)$, $s_2 = 2R \sin(2\pi/7)$, $s_3 = 2R \sin(3\pi/7)$ — one chord of each class — and with area

$$\Delta = \frac{s_1 s_2 s_3}{4R} = \frac{\sqrt{7}}{4} R^2 = 0.6614378277661476 \dots R^2.$$

Proof sketch. The vertices of the line $L_0 = \{0, 1, 3\}$ sit at central angles $0, 2\pi/7, 6\pi/7$. The inscribed-angle theorem gives the interior angles as half the opposite arcs: $\pi/7$ at the vertex $\{3\}$ -side, $2\pi/7$ and $4\pi/7$ — the binary ladder $1:2:4$, summing to π . The sides are the chords of arcs $2\pi/7, 4\pi/7, 8\pi/7$, i.e. $2R \sin(\pi/7)$, $2R \sin(2\pi/7)$, $2R \sin(4\pi/7) = 2R \sin(3\pi/7)$ — one chord of each class, by the sine symmetry

$\sin(4\pi/7) = \sin(3\pi/7)$. The three vertices lie on the circumcircle of radius R , which is therefore the triangle's circumcircle, and the area of any triangle is $abc/4R_{\text{circum}}$:

$$\Delta = \frac{2R \sin \frac{\pi}{7} \cdot 2R \sin \frac{2\pi}{7} \cdot 2R \sin \frac{3\pi}{7}}{4R} = 2R^2 \sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = 2R^2 \cdot \frac{\sqrt{7}}{8} = \frac{\sqrt{7}}{4} R^2,$$

by the classical identity $\sin(\pi/7)\sin(2\pi/7)\sin(3\pi/7) = \sqrt{7}/8$. Every line is a rotation image of L_0 , so the seven triangles are congruent. $\square$

The chord product carries the same cubic field: $s_1 s_2 s_3 = 8R^3 \cdot \sqrt{7}/8 = R^3 \sqrt{7}$. At the registered scale $R=1$ AU the figure's dimensions in the sky are:

quantity

closed form

value (30 digits)

side s_1

$$2 \sin(\pi/7)$$

0.867767478235116240951536743604

short diagonal s_2

$$2 \sin(2\pi/7)$$

1.56366296493605961741688916157

long diagonal s_3

$$2 \sin(3\pi/7)$$

1.94985582436364721403626316285

product $s_1 s_2 s_3$

$$\sqrt{7}$$

2.64575131106459059050161575194

cell area Δ

$$\sqrt{7}/4$$

0.661437827766147647625403937985

lattice rate ω_7

$3/2\pi$

0.477464829275686007306651290118

angle ladder

$\pi/7:2\pi/7:4\pi/7$

1:2:4

Stage P1 pins these literals at $\text{mp.dps} = 30$ and certifies the float64 layer against them to 4.9×10^{-16} ; stage P4 re-derives the congruence of all seven line-triangles to 6.7×10^{-16} absolute.

4. The Kepler register and the mass correction (Lemma D)

4.1 The committed register

The data layer commits the NASA register verbatim: the GM of the Sun ($1.32712440018 \times 10^{20} \text{ m}^3 \text{ s}^{-2}$) and of the eight planets, the equatorial radii, the semi-major axes and eccentricities (JPL J2000 mean elements), the sidereal periods and the surface gravities. The two-body constant of planet i is $\mu_i = 4\pi^2(1+m_i/M_\odot)$.

4.2 The dynamic certification

Lemma D. *The two-body period of every planet, measured from interpolated perihelion passages of the RK4 flow, matches $T_i = 2\pi\sqrt{a_i^3/\mu_i}$ to 6.2×10^{-13} relative; the uncorrected formula $2\pi\sqrt{a_i^3/4\pi^2}$ misses by the exact signature $\sqrt{1+m_i/M_\odot} - 1 = m_i/2M_\odot + O(m_i^2)$, up to 4.77×10^{-4} for Jupiter.*

Proof sketch. The two-body relative flow is exactly Keplerian, so the measured period converges to T_i at the integrator's phase accuracy — the parabolic interpolation of the perihelion minima cancels its bias between successive minima to $(ndt)^4$. The uncorrected ratio is $T_{\text{uncorr}}/T_i = \sqrt{\mu_i/4\pi^2} = \sqrt{1+m_i/M_\odot}$ exactly, whence the signature. $\square$

The recorded separation between the corrected and uncorrected registers is a factor 7.7×10^8 — the mass correction is not a refinement but the difference between physics and its spherical-cow approximation. The committed fact-sheet periods are recorded as a provenance diagnostic: they disagree with the Kepler-implied periods at up to 4.8×10^{-4} (Neptune worst) because they are rounded and epoch-mixed; the bench measures Kepler III in the dynamics instead of quoting the tables, and records the table inconsistency honestly.

5. The planetary simulation (stage P3)

The full system — the Sun and the eight planets, all pairwise Newtonian terms, barycentric momentum nulled exactly — is integrated for 12 years at $dt = 2 \times 10^{-4}$ yr (the quick preset: 4 years at 10^{-3} ; the full preset: 40 years at 10^{-4}). The committed initial condition places every planet at perihelion of its committed (a, e) orbit, all perihelia aligned, with the barycentric frame shift applied along the motion axis so that the heliocentric relative states are exact and the total momentum is machine-zero.

The recorded registers of the default run:

register	
recorded value	
registered band	
energy drift (12 yr)	
	2.07×10^{-13}
	$\leq 10^{-9}$
angular-momentum drift	
	0.0 (bit-identical)
	$\leq 10^{-10}$
worst osculating $\Delta a/a$	
	4.9×10^{-3}
	$\leq 10^{-2}$
worst osculating Δe	
	4.8×10^{-3}
	$\leq 10^{-2}$
Kepler III of the integrated motion	
	2.5×10^{-5}
	$\leq 10^{-4}$
perturbation hierarchy	
	2.2×10^{-3}
recorded	

The measured periods over integer revolutions — Mercury 87.968 d, Venus 224.689 d, Earth 365.243 d, Mars 686.917 d — differ from the committed table by 10^{-5} – 10^{-4} : the mutual perturbations plus the table's epoch mixing, both honest physics-and-provenance effects. The perturbation hierarchy record of 2.2×10^{-3} quantifies the classical smallness of the planetary problem: even the most perturbed body feels the rest of the system at two parts in a thousand of the solar pull.

6. The heptagon lattice (Theorem C)

Theorem C. *The seven-vortex regular heptagon ($\Gamma_i = \Gamma$, R) is a relative equilibrium rotating rigidly at*

$$\omega_7 = \frac{\Gamma(N-1)}{4\pi R^2} = \frac{3\Gamma}{2\pi R^2} = 0.4774648292756860\dots$$

at $N=7$; its co-rotating spectrum is Havelock-stable with $\max \operatorname{Re} \lambda$ on the numerically-zero floor (7.0×10^{-9} , classifier 10^{-7}); and its Hamiltonian is exactly

$$H_{ring} = -\frac{7\Gamma^2}{2\pi} \ln(\sqrt{7} R^3).$$

Proof sketch. The rotation formula is the roots-of-unity register of the sibling bench at $N=7$; the stability classification is the Havelock register — $N=7$ is the last stable level, $N=8$ the first unstable one. The Hamiltonian: the 21 pairwise distances of the ring are seven of each chord class, so $\sum_{i<j} \ln d_{ij} = 7 \ln s_1 + 7 \ln s_2 + 7 \ln s_3 = 7 \ln(R^3 \sqrt{7})$ by Theorem A's product identity, giving $H_{ring} = -\frac{7\Gamma^2}{2\pi} \ln(\sqrt{7} R^3) = 7 H_{cell}$. ◻

Stage P5 measures the rotation to 1.35×10^{-12} relative over two revolutions, holds the invariants to 1.25×10^{-14} , and records the exact-Hamiltonian identity at error 0.0 — the closed form -1.08395426662194 at $\Gamma = R = 1$ equals the numeric Kirchhoff register bit-for-bit in float64.

7. The seven cells (Theorem B)

7.1 The exact cell registers

Theorem B. *Place seven equal vortices at the heptagon vertices. Then every Fano line-cell carries the same Hamiltonian, angular impulse and moment of inertia,*

$H_{cell} = -\frac{\Gamma^2}{2\pi} \ln(\sqrt{7} R^3)$, $I_{cell} = 3\Gamma R^2$, *and the same rotation-invariant impulse magnitude $|P+iQ|$; the seven cells are congruent as dynamical systems, and their shape cycles are periodic with one common period.*

Proof sketch. Each line is a rotation image of L_0 , so the cells are congruent — equal invariants follow. The Hamiltonian of one cell is $-\frac{\Gamma^2}{2\pi}(\ln s_1 + \ln s_2 + \ln s_3) = -\frac{\Gamma^2}{2\pi} \ln(R^3 \sqrt{7})$ by Theorem A's product identity — the same for all seven. The moment of inertia is $\Gamma \sum |z_k|^2 = 3 \Gamma R^2$: every vertex sits on the circumcircle. The shape periodicity is the classical integrability of the three-vortex problem: the shape variable moves on the closed level curve of H inside the fixed- I shape domain. ▢

7.2 The measurement

Stage P6 integrates all seven cells concurrently (one batched RK4 over the $(7, 3, 2)$ position array) and records:

the seven Hamiltonians equal to 1.79×10^{-16} relative of the closed form, with spread 5.6×10^{-17} (absolute);

the invariant drifts along the shape cycle below 9.2×10^{-14} ;

the seven shape periods $T_{shape} = 9.3814\dots$ agreeing with spread **exactly 0.0** (the same integration, the same grid, congruent initial data);

the return residual 3.9×10^{-7} ;

the honest obstruction: the maximal shape departure is 0.288 — a scalene Fano cell is **not** a relative equilibrium, the shape genuinely leaves and returns.

The last register is the bench's honesty anchor: the figure does not pretend its cells are rigid. What PSL(2,7) buys is not rigidity of the cells but the *equality of their cycles* — a dynamical symmetry, not a static one.

8. The gravity bridge and the mass-ladder deficit (Lemma E)

8.1 Gravity as exact geometry

Stage P7 binds the committed register to exact closed forms in three registers.

The Schwarzschild ladder. $r_s = 2GM/c^2$: the Sun 2953.25 m, Jupiter 2.82 m, Saturn 0.84 m, Neptune 0.46 m, Uranus 0.39 m, Earth 8.87 mm, Venus 7.23 mm, Mars 0.95 mm, Mercury 0.49 mm. The arithmetic is certified against mpmath at 1.0×10^{-16} — gravity rendered as literal, exact figure dimensions, nine orders of magnitude from the Sun's horizon to Mercury's millimetre.

The Hill margins. For adjacent planet pairs the ratio $\Delta a / (r_{H,i} + r_{H,j})$ never drops below 5.08 (the pair Jupiter–Saturn) — the non-crossing certificate of the planetary cells, registered at ≥ 3 .

The figure dimensions in the sky. The chords s_1, s_2, s_3 of Theorem A at $R=1$ AU: 1.298×10^{11} m, 2.339×10^{11} m, 2.917×10^{11} m.

8.2 The deficit

Lemma E. *The gravity ladder $\delta_i = \log_{10}(G M_i / G M_{\oplus})$ summed over the seven Fano lines ranges from 0.008 (Mercury+Mars+Neptune) to 6.715 (Jupiter+Saturn+Neptune) — a spread of 6.71 dex; and a circulation ladder Γ_i built from the same weights deforms the heptagon lattice by 0.298 within one rotation while the equal-circulation ring stays rigid to 9.8×10^{-15} .*

The interpretation is the bench's headline honesty: the figure is mass-blind by construction — $PSL(2, 7)$ treats its seven points equally — while the solar system spans eight orders of magnitude in gravitational strength. The bench does not tune the figure to hide this; it measures the deficit twice, statically (the line sums) and dynamically (the weighted lattice), and records both. The mean-motion ladder of the planets spans 684.1 from Mercury to Neptune against the figure's chord ladder of 2.247 — recorded in the same diagnostics block: the figure is a register of symmetry, not a metric model of the sky.

9. The numerical protocol summary

stage
register
tolerance
recorded
wall (default)
P1
figure literals + Kepler–Earth anchor
 $10^{-15} / 10^{-4}$
 $4.9 \times 10^{-16} / 9.7 \times 10^{-6}$
0.2 s
P2
Kepler III, two-body, 8 planets
 10^{-9}
 6.2×10^{-13}
1.2 s
P3
N-body conservation + elements
 $10^{-9} / 10^{-2} / 10^{-4}$
 $2.1 \times 10^{-13} / 4.9 \times 10^{-3} / 2.5 \times 10^{-5}$
5.3 s
P4
PSL(2,7) algebra
exact / 10^{-12}
exact / 6.7×10^{-16}
0.04 s
P5
the lattice

$$10^{-9} / 10^{-12} / 10^{-7}$$

$$1.3 \times 10^{-12} / 1.2 \times 10^{-14} / 7.0 \times 10^{-9}$$

0.8 s

P6

the seven cells

$$10^{-12} / 10^{-6}$$

$$9.1 \times 10^{-14} / 0.0$$

7.7 s

P7

the bridge

$$10^{-12} / \geq 3$$

$$1.0 \times 10^{-16} / 5.08$$

0.4 s

X1

PSL(2,7) from first principles

exact integers

exact; simple

0.03 s

X2

the two natural actions + the Sylow census

exact integers

exact; bridge found

0.15 s

X3

the (2,3,7) generation + Hurwitz + Klein

exact integers

exact; 0 non-generating

1.6 s

X4

the hyperbolic $\{7,3\}$ figure + the disk witness

$$10^{-48} / 10^{-9}$$

$$3.2 \times 10^{-29}; \text{witness } 0.0 / 1.1 \times 10^{-16}$$

0.2 s

X5

the integrator certification

slopes ± 0.2 ; trend ≤ 0.1

$$2.0004 / 3.995; 0.006 \text{ vs } 0.99$$

3.3 s

X6

the 1PN GR bridge

$5 \times 10^{-3} / 5 \times 10^{-8}$

4.6×10^{-4} ; Mercury 42.982 "/century

7.0 s

Presets: quick (≈ 8 s smoke of both suites), default (≈ 33 s, the committed protocols), full (≈ 120 s, release validation: 40 years of N-body at $dt=10^{-4}$, 60-digit mpmath literals, four ring revolutions, 60-orbit 1PN windows). The protocols are deterministic; a git diff on results/protocols/ after a re-run must be empty (the run stamp excepted). The X-register records into the same envelope with suite = planetvortex-hardcore.

10. Honesty notes

The figure is a **register of symmetry, not a metric model of the sky**. No claim of physical vortex-planet identity is made; the direction of every quantitative statement is from the registers toward the figure, measuring the distance, never from the figure toward predictions about the planets.

The mass ladder **breaks** the figure's symmetry; the bench quantifies the deficit (6.71 dex static, 0.298 dynamic) instead of tuning it away.

The planetary simulation is a **planar idealization** with aligned perihelia; its registers certify the dynamics against the committed initial condition, not the calendar ephemeris.

The fact-sheet periods are **rounded and epoch-mixed** at 4.8×10^{-4} ; the Kepler register is dynamic, and the table inconsistency is recorded, not absorbed.

Stability is **spectral (linear)**; nothing beyond the linearized classification is claimed.

The three-vortex cells are **integrable but not rigid**; the scalene cell is not a relative equilibrium, and the shape cycles are the honest substitute.

The measured planetary periods differ from the committed table at 10^{-5} – 10^{-4} — mutual perturbations plus provenance, quoted as physics, not dismissed as error.

11. Резюме (RU)

Мини-репозиторий planetvortex строит «гравитационную фигуру» Солнечной системы: правильный семиугольник радиуса $R=1$ а.е. с Солнцем в центре и семью планетами-блуждающими в вершинах (Меркурий, Венера, Марс, Юпитер, Сатурн, Уран, Нептун — в порядке периодов), на котором начерчены семь прямых Фано $\{i, i+1, i+3\}$ — точечно-линейная структура с группой автоморфизмов $PSL(2, 7)$ порядка 168.

Теорема А вычисляет точные размеры фигуры: каждая прямая Фано — конгруэнтный разносторонний треугольник с углами $(\pi/7, 2\pi/7, 4\pi/7)$, сторонами $2R \sin(k\pi/7)$ и площадью $\sqrt{7}R^2/4$ — по классическому тождеству произведения синусов. Теорема В показывает, что семь трёхвихревых ячеек Тривортекса несут в точности одинаковые гамильтонианы $-(\Gamma^2/2\pi) \ln(\sqrt{7}R^3)$ и конгруэнтные шейп-циклы с периодом 9,3814 и разбросом ровно ноль — динамическая тень $PSL(2,7)$. Теорема С сертифицирует семиугольную решётку как устойчивое по Хавелоку относительное равновесие с $\omega_7 = 3\Gamma/2\pi R^2$ — строка $N=7$ скана соседнего стенда, запинненная бит-в-бит. Лемма D сертифицирует третий закон Кеплера динамически: с поправкой на массу — до $6,2 \times 10^{-13}$ по всем восьми планетам, без неё — промах на физическую сигнатуру $m_i/2M_\odot$ до $4,77 \times 10^{-4}$ (Юпитер). Лемма E честно измеряет дистанцию между фигурой и небом: лестница гравитаций даёт разброс сумм по прямым 6,71 dex, а мягкая гравитационно-взвешенная лестница циркуляций деформирует вихревую фигуру на 0,30 за одно вращение при жёсткости равноциркуляционного кольца до $9,8 \times 10^{-15}$.

Полное ньютоновское моделирование «Солнце + 8 планет» (12 лет, RK4) даёт дрейф энергии $2,1 \times 10^{-13}$; все семь ступеней P-лестницы закоммичены как детерминированные JSON-протоколы, и каждый протокол перевыверяется CI на каждый пуш. Фигура — регистр симметрии, а не метрическая модель неба: стенд измеряет дефицит связи, а не прячет его.

Appendix X — the hardcore register (X1–X6)

The P-ladder registers; the X-register attacks. Where a ladder stage asserts a claim with a committed tolerance, the corresponding X stage re-derives the same claim by a second, independent route — brute-force enumeration where the objects are finite, exact integer arithmetic where the statement is arithmetic, and a numerical construction where a closed form could be quietly wrong. The appendix records the six stages; the committed protocols carry the full parameter snapshots.

X1 — the group from nothing. All $49^2 = 2401$ matrices over F_7 are enumerated and classified: 2016 invertible (exactly $(7^2 - 1)(7^2 - 7)$), 336 of determinant one (exactly $7(7^2 - 1)$), and — quotienting by $\pm I$ — 168 classes, each with exactly two lifts. The centre of $SL(2,7)$ is $\{\pm I\}$; the centre of the quotient is trivial. The conjugacy computation partitions the group into classes of sizes 1, 21, 24, 24, 42, 56 — the classical class equation — and the element-order census reads $1 + 21 + 56 + 42 + 48 = 168$. Simplicity is certified exhaustively: every normal subgroup is a union of conjugacy classes containing the identity, and all $2^5 = 32$ such unions are checked against Lagrange — none of the intermediate sizes divides 168.

X2 — the two actions. Conjugation on the Sylow subgroups yields the two natural permutation representations. The Sylow census is computed, not quoted: $n_7=8$ (the projective line $P^1(F_7)$), $n_3=28$, $n_2=21$ with the self-normalizing dihedral Sylow-2 of order 8. The 8-point action is certified 2-transitive (one orbit on the 56 ordered pairs) and 3-homogeneous (one orbit on the 56 triples); the point stabilizer is the Frobenius group of order 21 with the exact census $(1, 14, 6)$. The 7-point action lives on one of the two conjugacy classes of S_4 subgroups (both are computed, 7+14 total counting the second class); the stabilizer carries the signature $(1, 9, 8, 6)$ of S_4 . The bridge: a permutation $\sigma \in S_7$ exists conjugating the action image onto the research model of fano.py — the Fano points of the figure are *proven* to be the seven conjugate S_4 subgroups.

X3 — the triangle generation and the Hurwitz arithmetic. Every one of the 336 pairs (a, b) with $|a|=2$, $|b|=3$, $|ab|=7$ generates the whole group (the product-order distribution over all $21 \times 56 = 1176$ pairs reads $\{2:168, 3:336, 4:336, 7:336\}$; a subgroup carrying orders 2, 3, 7 has order divisible by 42, and orders 42 and 84 are excluded by the coset action and simplicity). The Klein relations $a^2=b^3=(ab)^7=[a, b]^4=1$ hold on the witness. The Hurwitz bound $84(g-1)=168$ and the Riemann–Hurwitz arithmetic $42(2g-2)=168$ are exact integers; the orbifold curvature of the $(2, 3, 7)$ signature is $-1/42$. The Klein quartic $x^3y+y^3z+z^3x=0$ is proven smooth without any symbolic dependency: multiplying the three gradient equations gives $27(xyz)^3=-(xyz)^3$, hence $28(xyz)^3=0$ — impossible over $\mathbb{C}$; every coordinate-zero branch collapses to the trivial vector. A smooth plane quartic has genus $(4-1)(4-2)/2=3$.

X4 — the hyperbolic figure and its witness. The fundamental right triangle of the $\{7, 3\}$ tiling (angles $\pi/7$ at the heptagon centre, $\pi/3$ at the vertex, $\pi/2$ at the edge midpoint) has the exact dimensions $\cosh(\ell/2)=\cos(\pi/7)/\sin(\pi/3)$, $\cosh r=\cos(\pi/3)/\sin(\pi/7)$, $\cosh R=\cot(\pi/7)\cot(\pi/3)$, verified against the hyperbolic Pythagoras $\cosh R=\cosh(\ell/2)\cosh r$ at 50 dps (residual 3.2×10^{-29} , the rounding floor of π itself). The area ladder is exact: triangle $\pi/42$, heptagon $\pi/3$, total $8\pi=2\pi(2g-2)$ — Gauss–Bonnet on the genus-3 surface. The combinatorial closure $3V=7F=2E=168$ and $14F=4E=6V=336$ holds with $V=56$, $E=84$, $F=24$, $\chi=-4$. As the independent witness, a regular heptagon is constructed directly in the Poincaré disk and its interior angle is solved to $2\pi/3$ by bisection: the resulting circumradius matches the closed form to 0.0 and the edge to 1.1×10^{-16} — two routes, one number.

X5 — the integrators certified. The convergence order is *measured*, not asserted: regressing $\log(\text{error})$ on $\log(dt)$ over the committed grids gives slope 2.0004 for the leapfrog and 3.995 for the Yoshida composition on a Kepler orbit with $e=0.4$. Forward–backward integration recovers the initial state to 7.5×10^{-12} (self-adjointness at roundoff). The symplectic energy envelope is bounded with no secular trend — the linear-trend ratio over an 80-period window is 0.006 (leapfrog) and 0.0002 (Yoshida) against 0.99 for the RK4 control on the same window — the drift signature that separates the two families.

Along a Yoshida orbit the Laplace–Runge–Lenz vector drifts at 5.5×10^{-11} , the vis-viva law holds to 2.3×10^{-12} and the exact orbit equation $r = p/(1 + e \cos \theta)$ to 7.3×10^{-11} .

X6 — the GR bridge. The 1PN acceleration (Schwarzschild test particle, harmonic gauge) is integrated for Mercury, Venus and Earth; the perihelion advance is measured from interpolated perihelion passages and compared with the closed form

$\Delta\varpi = 6\pi GM/(a(1 - e^2)c^2)$. The worst measured-over-formula deviation is 4.6×10^{-4} ; the Newtonian control run advances 1.9×10^{-10} rad/orbit — integrator zero, as it must before the PN measurement means anything. Mercury’s precession integrates to 42.982 " per century against the textbook 42.98 " (relative error 5.3×10^{-5}), computed from the same committed register the figure binds; the full eight-planet formula ladder descends monotonically from 42.98 " to 0.0008 ".

The layer adds no physics and removes none: it is the same bench attacked from the sides the P-ladder leaves open. Its protocols live in the same folder, under the same envelope, behind the same discipline — every tolerance committed before the recorded run.

References

Trivortex — the parent framework: Theorem 3.1, the V1–V4 ladder, the mini-research program. Repository root README, DOI 10.5281/zenodo.21825394.

Polyvortex — the N-vortex extension bench: the W-ladder, the Havelock threshold at $N=7 \vee 8$, the $\pi \ln 2$ admissibility boundary. polyvortex/ of the same repository.

Cycloring — the ring laboratory: the Gamma-period registers and the W-ladder. cycloring/ of the same repository.

NASA NSSDC planetary fact sheets — the GM, radii, periods and surface gravities of the committed register. <https://nssdc.gsfc.nasa.gov/planetary/factsheet/>

JPL, Keplerian elements for approximate positions of the major planets — the J2000 mean elements of the committed register. https://ssd.jpl.nasa.gov/planets/approx_pos.html

H. Aref, Motion of three vortices — the classical integrability and the shape periodicity of the three-vortex problem.

J. J. Thomson, A Treatise on the Motion of Vortex Rings — the N-vortex ring and its relative equilibria.

H. S. M. Coxeter, Twelve Geometric Essays — the Fano plane, the Klein quartic and the group of order 168.